

Module 1 Assessment

1) (a) Problem that arises from maintaining siloed data systems

- Data silo is where isolated systems or departments have different databases restricted from others

(b) How do these challenges impact business operations and decision making

- Since data is isolated, relevant linked data which is essential for decision making such as ~~data~~ customer data and supply chain information

2 (a) A scalable Data Integration system.

Solution $\Rightarrow$ A modern Hybrid data Architecture

Data source Layer - Legacy database, and cloud system

Integration layer - API gateway

Central data platform - Such as data lake and warehouse

Data Access Layer - Dashboards and Analytics platform

(b) Technologies and Tools

Tools for Data storage - Warehouses, Lakehouses, Database.

Streaming - Apache Kafka.

Integration tools - API management and ETL

3) a) Strategies of maintaining data quality and consistency

- Data standardization - rules
- Tracking data origin and transformation
- Automated checks during data ingestion

b) ~~Metric to monitor the status of~~
Plan to ensure compliance with data pr

- Data Mapping - Data Classification
- Audit trails - Access control.

4) Steps of implementing data quality and consistency during integration process

- Assessment - Auditing data sources
- Design - Define Architecture and choose tools and platform
- Development - Building ELT and ETL pipeline
- Testing
- Deployment
- Training

5) Metric KPI's

- i) Data Accuracy
- ii) Duplicate rate
- iii) Data Latency
- iv) System Uptime